A Master Quadratic with Twelve Equivalent Presentations: Special-Function Factorizations and a High-Precision Empirical Ansatz for α^{-1}

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Abstract

We study the monic quadratic

$$P(x) = x^2 - 16(G^*)^2\, x + 16(G^*)^3, \tag{1}$$

where the dimensionless constant G^* admits multiple equivalent representations in terms of classical special functions associated with the lemniscatic modulus $k = 1/\sqrt{2}$. We give explicit algebraic proofs that the "Forms 1–8" listed below are exact rewritings of (1), and we organize the resulting invariants (Vieta data, discriminant, modular parameter, nome, etc.) in a single framework.

Separately, we record an empirical correction ansatz involving $\varepsilon = e^\pi - \pi - 20$ that maps the larger root x_+ to the CODATA 2022 inverse fine-structure constant α^{-1} at $\sim 2 \times 10^{-13}$ relative error using only low-order powers of $|\varepsilon|$. We emphasize a strict taxonomy: the equivalences among the special-function forms are theorems; the identification of x_+ and the correction ansatz with α^{-1} are testable hypotheses that require independent physical justification.

1 Definitions and the master quadratic

Definition 1 (Lemniscatic constants)**.** *Let Γ denote Euler's gamma function. Define*

$$G^* \coloneqq \frac{\sqrt{2}\,\Gamma(1/4)^2}{2\pi}, \tag{2}$$

$$K \coloneqq K\left(\frac{1}{\sqrt{2}}\right), \qquad \omega \coloneqq \frac{\Gamma(1/4)^2}{4\sqrt{\pi}}, \tag{3}$$

$$I \coloneqq \int_0^1 \frac{dt}{\sqrt{1-t^4}}, \qquad M \coloneqq \mathrm{AGM}\left(1, \frac{1}{\sqrt{2}}\right), \tag{4}$$

$$\tau \coloneqq i, \qquad q \coloneqq e^{i\pi\tau} = e^{-\pi}. \tag{5}$$

Here $K(k)$ is Legendre's complete elliptic integral of the first kind and AGM *is the arithmetic–geometric mean.*

With (2), the master quadratic is (1). Since it is monic, its two roots satisfy Vieta invariants

$$x_+ + x_- = 16(G^*)^2, \qquad x_+x_- = 16(G^*)^3,$$

and the discriminant is $\Delta = (16(G^*)^2)^2 - 64(G^*)^3$.

Numerical anchor (for reproducibility)

Using high-precision arithmetic,

$$G^* \approx 2.95867511918863889, \quad x_+ \approx 137.036171458155484, \quad x_- \approx 3.023963916339021.$$

2 The twelve forms and why each matters

Table 1 records the twelve proposed presentations, grouping them into (i) *exact* rewritings (Forms 1–8), (ii) modular-data invariants (Forms 10–11), and (iii) an *empirical* correction map (Form 12).

#	Formula	Why it is significant
1	$x^2 - 16(G^*)^2 x + 16(G^*)^3 = 0$	Canonical monic quadratic; fixes all polynomial invariants.
2	$x^2 - \frac{8\Gamma(1/4)^4}{\pi^2}x + \frac{4\sqrt{2}\,\Gamma(1/4)^6}{\pi^3} = 0$	Expresses coefficients via $\Gamma(1/4)$ (lemniscatic special value).
3	$x^2 - \frac{128K^4}{\pi}x + \frac{64\sqrt{2}\,K^6}{\pi^{3/2}} = 0$	Converts to elliptic integral $K(1/\sqrt{2})$: modular geometry input.
4	$x^2 - \frac{64\omega^4}{\pi}x + \frac{64\sqrt{2}\,\omega^6}{\pi^{3/2}} = 0$	Lattice/period notation: ω is a quarter-period scale.
5	$x^2 - \frac{128I^4}{\pi}x + \frac{64\sqrt{2}\,I^6}{\pi^{3/2}} = 0$	Pure definite integral form; useful for asymptotics and bounds.
6	$x^2 - \frac{4\pi^2}{M^4}x + \frac{2\sqrt{2}\,\pi^3}{M^6} = 0$	Iterative dynamics via AGM; yields fast computation and rigidity.
7	$x^2 - 16\pi^2\theta_3(0,e^{-\pi})^8 x + 16\sqrt{2}\,\pi^3\theta_3(0,e^{-\pi})^{12} = 0$	Theta/nome form at $\tau = i$: makes modular structure explicit.
8	$x^2 - 8\pi^2\theta^2 x + 8\pi^3\theta^3 = 0$	Introduces a compact parameter θ; convenient for rescalings.
9	$x^2 - 4k_c^2 x + \frac{4k_c^3}{G^*} = 0, \quad k_c = \frac{4}{G^*}$	Mandelbrot/cardioid bridge via $k_c c_{cusp} G^* = 1$ with $c_{cusp} = 1/4$; suggests dynamical-system normal forms.
10	$\tau = i$	Self-dual modular point; canonical CM fixed point.
11	$j(i) = 1728$	CM invariant controlling elliptic curve isomorphism class.
12	$\alpha^{-1} \stackrel{?}{=} x_+ - \frac{9}{47}\lvert\varepsilon\rvert + \frac{5}{64}\lvert\varepsilon\rvert^2 + O(\varepsilon^3), \ \ \varepsilon = e^\pi - \pi - 20$	Empirical map to a physical constant; falsifiable and coefficient-sensitive.

Table 1: The twelve presentations (Forms 1–12) and their roles. Forms 1–8 are exact algebraic rewritings once identities for special values are accepted; Forms 10–11 are modular invariants; Form 12 is an empirical ansatz.

3 Exact equivalence of Forms 1–8

Theorem 1 (Equivalence chain)**.** *Assuming the standard special-value identities for the lemniscatic modulus $k = 1/\sqrt{2}$, Forms 1–8 are mathematically equivalent: each is obtained from the previous by substitution of definitions (2)–(4) and standard identities relating K, ω, I, M, and θ_3.*

Proof sketch. Form 2 is immediate from eliminating G^* via (2). Form 3 uses $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$. Form 4 is $\omega \equiv K(1/\sqrt{2})$. Form 5 is $I = \omega/\sqrt{2}$ (equivalently, $\omega = \sqrt{2}\,I$). Form 6 uses $K(k) = \pi/(2\,\mathrm{AGM}(1,\sqrt{1-k^2}))$ with $k = 1/\sqrt{2}$. Form 7 uses the standard nome–theta representation $K(k) = \frac{\pi}{2}\theta_3(0|\tau)^2$ at $\tau = i$ (thus $q = e^{-\pi}$). Form 8 is a notational compression defining θ proportional to $\theta_3(0, e^{-\pi})^{-4}$. Each step is a direct substitution into Form 1. □

4 Invariants of the master quadratic

From the polynomial viewpoint, the primitive invariants are $(S, P) = (16(G^*)^2, 16(G^*)^3)$ and the discriminant $\Delta = S^2 - 4P$. From the modular viewpoint, the invariants are $(\tau, q, j(\tau)) = (i, e^{-\pi}, 1728)$ and the associated lemniscatic special values of K and θ_3.

5 Form 12: a correction ansatz for α^{-1}

Definition 2 (The epsilon offset)**.** *Define*

$$\varepsilon := e^{\pi} - \pi - 20. \tag{6}$$

Numerically $\varepsilon \approx -0.000900020810524232734$.

Proposition 1 (Low-order correction map)**.** *Define*

$$\alpha_{\text{ans}}^{-1} \;:=\; x_+ - \frac{9}{47}|\varepsilon| + \frac{5}{64}|\varepsilon|^2. \tag{7}$$

Then

$$\alpha_{\text{ans}}^{-1} \approx 137.035999177029134,$$

and compared to the CODATA 2022 central value $\alpha^{-1} = 137.035999177$*, the absolute and relative deviations are*

$$|\Delta\alpha^{-1}| \approx 2.9134478498585 \times 10^{-11}, \qquad \left|\frac{\Delta\alpha^{-1}}{\alpha^{-1}}\right| \approx 2.1260456138211 \times 10^{-13}.$$

Remark 1 (Taxonomy)**.** *The arithmetic agreement in Proposition 1 is* not *a derivation of α; it is a numerical fit unless and until the coefficients and the appearance of ε are derived from a physical principle. What the fit* is*: a compact, falsifiable hypothesis whose failure/success can be checked against future independent determinations of α.*

6 A "Casimir gap" proxy from the second root

The smaller root is close to 3:

$$x_- \approx 3.023963916339021.$$

If one identifies 3 as a baseline integer (e.g. an $SU(3)$ color-count proxy), then the deviation

$$\delta_C := x_- - 3 \approx 0.023963916339021 \tag{8}$$

is a natural dimensionless "gap" parameter. Any Casimir-like interpretation requires a concrete model that turns a dimensionless gap into a measurable scale.

7 Five concrete physics embedding programs (speculative)

The exact mathematics supplies a rigid *special-function skeleton*. To connect it to physics without handwaving, one must propose a mapping from that skeleton into a dynamical or kinematical structure. Below are five concrete programs; each becomes meaningful only when a Lagrangian/Hamiltonian, symmetry principle, or renormalization statement is specified.

1. **RG fixed point at $\tau = i$.** Use $\tau = i$ (self-dual under $\tau \mapsto -1/\tau$) as a renormalization-group fixed point and interpret $q = e^{-\pi}$ as an instanton/finite-size suppression factor. The master quadratic then defines a candidate dimensionless coupling built from (K, θ_3, q).

2. **Spectral quantization via AGM.** Treat $M = \mathrm{AGM}(1, 1/\sqrt{2})$ as the value of a convergent discrete iteration and map it to a discrete coarse-graining flow. Observables are then functions of iteration invariants; "quantization" arises from the iteration's attractor structure.

3. **Vacuum-energy proxy from δ_C.** Interpret $x_- = 3 + \delta_C$ as "integer + vacuum shift" and attempt to embed δ_C as a renormalized Casimir coefficient (dimensionless) multiplying a geometry-dependent scale (plate separation, curvature radius, compactification length).

4. **Fractal dynamics bridge.** Form 9 suggests a normalization to a cusp constant 1/4. A concrete embedding would interpret this as a dynamical-system normal form (e.g. parabolic cusp scaling) whose coefficients are forced to be lemniscatic.

5. **Two-sector split.** Treat (x_+, x_-) as two coupled sectors (e.g. "ontic/epistemic") whose interaction is encoded by the fixed Vieta invariants. A testable model must specify how couplings deform observables while preserving the polynomial invariants.

A Reproducibility code

```
import mpmath as mp
mp.mp.dps = 80
pi = mp.pi
G = mp.sqrt(2)*mp.gamma(mp.mpf(1)/4)**2/(2*pi)

# master quadratic roots:
b = -16*(G**2); c = 16*(G**3)
disc = b*b - 4*c
x_plus  = (-b + mp.sqrt(disc))/2
x_minus = (-b - mp.sqrt(disc))/2

eps = mp.e**pi - pi - 20
alpha_ans_inv = x_plus - (mp.mpf(9)/47)*abs(eps) + (mp.mpf(5)/64)*abs(eps)**2
print(G); print(x_plus); print(x_minus); print(alpha_ans_inv)
```

References

References

[1] NIST Digital Library of Mathematical Functions. Chapters 19 and 20 (Elliptic Integrals and Theta Functions). National Institute of Standards and Technology. https://dlmf.nist.gov/

[2] CODATA Recommended Values of the Fundamental Physical Constants (2022). National Institute of Standards and Technology (NIST). https://physics.nist.gov/cuu/Constants/

[3] For the modular invariant $j(i) = 1728$ and complex multiplication background, see standard references on elliptic curves and modular functions (and encyclopedic summaries such as https://en.wikipedia.org/wiki/J-invariant).